DaSEH Day 4 Cheatsheet

Data Summarization

Functions

Library/Package	Piece of code	Example of usage	What it does
Base R	`min()`	`min(x)`	Returns the minimum value in object `x`.
Base R	`sum()`	`sum(x)`	Returns the sum of values in object `x` (integer, numeric, or logical).
Base R	`mean()`	`mean(x)`	Returns the arithmetic mean of values in object `x` (integer, numeric, or logical).
Base R	`log()`	`log(x)`	Returns the natural logarithm of object `x`. Use `log2(x)` for base 2, or specify another base.
Base R	`range()`	`range(x)`	Returns the minimum and maximum values in object `x`.
Base R	`sd()`	`sd(x)`	Returns the standard deviation of object `x`.
Base R	`sqrt()`	`sqrt(x)`	Returns the square root of object `x`.
Base R	`quantile()`	`quantile(x, probs = .5)`	Returns sample quantiles for specified probabilities.
Base R	`summary()`	`summary(x)`	Returns a summary of values in object `x`.
`tidyverse` (`dplyr`)	`pull()`	`x_vect <- df |> pull(x)`	Extracts a single column as a vector. Useful before summary functions like `mean()` and `sum()`.

Library/Package	Piece of code	Example of usage	What it does
`tidyverse` (`dplyr`)	`summarize()`	`df <- df |> summarize(mean_x = mean(x))`	Summarizes values into one or more output values. `summarize()` and `summarise()` are synonyms.
`tidyverse` (`dplyr`)	`distinct()`	`df |> distinct(factor_name)`	Displays unique rows from a data frame or tibble.
`tidyverse` (`dplyr`)	`n_distinct()`	`x_vect |> n_distinct()`	Counts unique values or combinations in one or more vectors.
`tidyverse` (`dplyr`)	`count()`	`df |> count(factor_name)`	Counts the number of rows in each group of a factor variable.
`tidyverse` (`dplyr`)	`group_by()`	`df |> group_by(factor_name)`	Groups rows by specified value(s).
Base R	`unique()`	`unique(df)`	Returns object `x` with duplicate elements or rows removed.
Base R	`rowSums()`	`rowSums(df)`	Calculates sums for each row.
Base R	`colSums()`	`colSums(df)`	Calculates sums for each column.
Base R	`rowMeans()`	`rowMeans(df)`	Calculates means for each row.
Base R	`colMeans()`	`colMeans(df)`	Calculates means for each column.

NOTE: Many summarizing functions (e.g., `mean()`, `sum()`) have the argument `na.rm = TRUE` to ignore missing data.

Data Classes

Major concepts

- **Character:** Strings or individual characters, quoted.
- **Numeric:** Any real number(s).
- **Double:** A subset of numeric values that can contain decimals.
- **Integer:** Whole number(s).
- **Factor:** Categorical or qualitative variables.
- **Logical:** Values composed of `TRUE` or `FALSE`.
- **Date/POSIXct:** Calendar dates and times.
- **Matrix:** Two-dimensional data where all rows and columns have the same data type.
- **Data frame:** Two-dimensional data where columns can have different data types.
- **List:** Data that can vary in dimensions and contain many data types, including vectors, strings, matrices, models, and other lists.

Functions

Library/Package	Piece of code	Example of usage	What it does
Base R	`class()`	`class(x)`	Returns the class of an object.
Base R	`as.numeric()`	`as.numeric(x)`	Coerces object `x` to numeric class. Similar functions include `as.character()`, `as.data.frame()`, `as.matrix()`, and `as.Date()`. Use with `mutate()`.
`tidyverse` (`lubridate`)	`ymd()`	`ymd("2024-01-31")`	Coerces character object `x` to date class. Use functions such as `mdy()` or `dmy()` for other date formats.

NOTE: `lubridate` is a powerful, widely used R package from `tidyverse` family to work with Date / POSIXct class objects.

* This cheatsheet format was adapted from Alex's Lemonade Stand materials (source).

* Find more resources at https://daseh.org.